Activity 6 Report: Database Backup, Restoration, and Recovery

COMP 018 Database Administration | Project tested: PUP-SPOT Campus Lost and Found (PHP + PostgreSQL) | Date of testing: 6 November 2026

1. Cover and links

Item	Value
Testing group	G08: Borreta, Joshua; Bola, Martin Lawrence; Juatchon, Joaquin Phillip; Marcos, Karl Raphael; Pacheco Jr., Sancho; Vera Cruz, Euan Joseph
Project tested	PUP-SPOT Campus Lost and Found
Owning group	Group 05 (grp05-dba)
Repository	https://github.com/grp05-dba/grp05_lostandfound
Tracker	https://github.com/EclipseChan/bwisIT-Solutions---Campus-Canteen-Pre-Order-and-Queue-Management-System/tree/main/other_group_tests
Evidence files	a6_backup.txt, a6_restore.txt, a6_verify.txt, a6_access.txt, a6_recovery.txt

Run on 6 November 2026 at 23:22 (Asia/Manila), after the application checks of Activity 3. The original database and the dump were never modified.

2. Backup plan

Item	Plan
Objects	Whole database grp05_lostandfounddb: 8 tables, 8 sequences, constraints and grants
Method	Logical backup with pg_dump custom format (-Fc)
Account	The project defines no backup role, so the dump was taken as postgres. Recommend a role with pg_read_all_data
Files outside the database	Uploaded photos are stored as files in the storage folder and the database keeps only their keys. The storage folder must be copied with every dump. The .env file with the credentials must be kept separately
Schedule (real deployment)	Daily full dump plus a dump before each deployment
Retention	7 daily and 4 weekly copies on a second machine
RPO and RTO	RPO 24 hours, RTO 1 hour
Roles	Roles are cluster-wide and not in a database dump. Keep a roles-only dump (pg_dumpall --roles-only)
Sensitive content	The dump holds names, student numbers, emails and phone numbers, so it must be stored with restricted access

3. Backup creation

```
pg_dump -U postgres -h localhost -Fc -d grp05_lostandfounddb -f
grp05_lostandfounddb_activity6.dump
```

Item	Value
File	grp05_lostandfounddb_activity6.dump
Size	32,805 bytes
SHA-256	0ED9D80E50D1C65127570A2EB79F32FFD784AE35EC40D239979F9A831CC26929
Archive	custom format, gzip, 89 TOC entries, 8 TABLE DATA entries, dumped by pg_dump 18.6
Errors	None, exit code 0

4. Restoration and verification

```
createdb -U postgres -h localhost grp05_lostandfounddb_restore
pg_restore -U postgres -h localhost -d grp05_lostandfounddb_restore
grp05_lostandfounddb_activity6.dump
```

Restore succeeded: exit code 0 in 0.2 seconds, no warnings or errors.

Check	Original	Restored	Match
Tables, views, sequences	8, 0, 8	8, 0, 8	Yes
Row counts (categories, claim_images, claims, item_images, items, lost_report_images, lost_reports, potential_matches)	11, 0, 4, 0, 7, 1, 9, 7	same	Yes
Constraints: check, foreign key, not null, primary key, unique	8, 8, 27, 8, 2	same	Yes
Table grants	97	97	Yes
Triggers	0	0	Yes
Application connects to the restored database	not required	not tested	n/a

5. Access control after restore

Table and column grants came back with the dump. The database-level privileges did not: the restored database had an empty datacl until CONNECT was granted to lostandfound_public, after which it matched the original. Logging in to the restored copy, lostandfound_public could read 7 items and lostandfound_app could read 4 claims, the same as on the original.

6. Recovery simulation

Step	Action	Result
Before	Checksum of claims in the restored copy	4 rows, md5 c1bb415b36f75e95457c3ceb5597b33d
Incident	DELETE of one claim row in the restored copy	DELETE 1, 3 rows, md5 e63cbd3ea16e8a4e3b6ad690703539b8
Recovery	createdb …_recovery and pg_restore of the same dump	Success, exit code 0
Compare	Checksum of claims in the recovery database	4 rows, same md5 c1bb415b…
Original untouched	Checksum in the original database	4 rows, same md5

Outcome: Recovered. A full restore into a fresh database was chosen because it leaves the damaged copy untouched for comparison. Data written after the last backup would be lost, which is the RPO.

7. Evaluation and recommendations

- Daily dumps are enough for a campus lost and found desk. A shorter RPO would need WAL archiving.
- The restore took 0.2 seconds, far inside the 1-hour RTO. Measure again as data grows.
- Copy the storage folder of photos with every dump, because a restore alone would leave records that point to missing files.
- Keep a roles-only dump and the grant script with each backup and re-apply the database-level CONNECT privilege after each restore.
- Create a dedicated backup role and protect the dump files because they contain personal data.

8. Limitations and errors

- Logical dumps capture one point in time only.
- The restore used the administrator account, so restored objects belong to postgres, the same as the original.
- No errors or warnings occurred. The only difference after restore was the missing database-level privileges.
- The application and the photo folder were not restored or pointed at the restored database. The test databases stay until the evidence is captured and are then dropped.

9. Feedback sent to owners

Sent on 6 November 2026: the dump and restore worked without warnings and all objects and row counts matched. The photo folder is outside the database and the database-level CONNECT privilege is not restored. The project has no backup role.

AI usage declaration

- Tool: Claude (Anthropic).
- Tasks: interpreting the results of the testing.
- Validation: the commands were run by the group on its own server, and every result in this report comes from the saved output files.